FPGA based Hardware Scheduling approach for optimising DSP pipelines

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This thesis is presented as part of the requirements for the conferral of the degree:

Bachelor of Software Engineering (Hons.)

Supervisor:
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The University of Sydney
School of Engineering

2025

Declaration

I, *Connor Sinclair*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Software Engineering (Hons.)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

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August 21, 2025

Abstract

Field-programmable gate arrays (FPGAs) offer unprecedented parallelism and low latency for real-time audio processing. However, FPGA toolchains typically perform static compilation and mapping of fixed designs. In practice, audio signals and user-defined module graphs vary over time, presenting an opportunity to predictively reconfigure the FPGA fabric. In this thesis, I propose an operator-theoretic and state-space analysis framework that models DSP modules as dynamical systems, using observables and custom signal correlation metrics to forecast workload and adjust resource allocation. Unlike conventional flows, this approach can anticipate the evolving demands of a modular audio effect chain (including nonlinear modules) and remap kernels (e.g. heavy multipliers or trigonometric units) onto LUTs, DSP slices, or BRAMs in accordance to resource demands. Expected contributions include: (1) an analytical model of audio DSP modules grounded in operator theory (Koopman embeddings) and state-space representations; (2) coherent metrics of signal interaction for resource tuning; and (3) a prototype toolchain enabling runtime FPGA optimization in an accessible form. This work bridges a gap between control-theoretic modeling and FPGA architecture, offering audio engineers and embedded developers a more adaptive FPGA DSP platform that can improve throughput, support complex nonlinear & stochastic audio effects and enable massive real-time parallelisation.

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Your first appendix

A.1 The title of the first section

The appendices work exactly the same way as chapters, they are numbered with letters rather than numbers though.